Contextual Provenance Defense for Persistent Memory Poisoning in LLM Agents

Research Proposal in Agentic Memory Systems

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Methods in Business Research (MG 3010), Assignment 1

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26 September 2026

Abstract Large language model agents use persistent memory to carry preferences, decisions and context across sessions, and that memory can also preserve an attack. An instruction hidden in an email, document, website or tool output may be stored today and change an answer or tool call weeks later. Recent studies show that such attacks can be split across several harmless-looking records or stay dormant until a later trigger, and that existing defenses judge memory too early, too expensively or without knowing where it came from. We propose Contextual Provenance Memory Guard (CPMG), a compact transformer cross-encoder that scores a memory together with its source, the current query or planned action, related memories and authenticated origin, at both write and retrieval time, alongside a deterministic policy that stops low-authority memory from authorising protected actions. The study is experimental. MPBench, extended with controlled compositional and dormant cases, is the main dataset, and CPMG will be compared with three recent baselines, PIGuard, A-MemGuard and an adapted BlindGuard, on attack success, low-false-positive detection, clean memory utility and cost.

Keywords agentic memory, memory poisoning, continual learning, contrastive learning, provenance, LLM agents

1 Introduction and Background

Persistent memory turns a stateless chatbot into an agent that can learn from earlier interactions. The agent observes a conversation or tool result, decides what to keep, embeds the saved record and later retrieves related records to answer or act. MemGPT showed that explicit memory tiers improve continuity across sessions [1], and benchmarks such as LoCoMo [11] and LongMemEval [12] now measure how well assistants remember over very long conversations. Memory has therefore become a standard part of agent design.

The same pipeline can amplify untrusted input. AgentPoison and MINJA showed that a small amount of poisoned memory can steer an agent while leaving normal accuracy almost unchanged [2], [3], and MPBench reported average attack success of about 50% across enterprise channels such as email, Slack, files and calendars [7]. A poisoned email may be reduced to a clean-looking summary, an instruction may be split across several memories, or a harmless record may wait for a later query to activate it.

The problem matters most for agents that act on a user's behalf. Our motivating case is Raymond, an executive Chief of Staff agent that reads email, Slack, documents, websites and tool outputs and reuses them during later planning. A website may supply a useful fact but should never approve a payment, and an email may suggest a meeting but should not silently overwrite a preference the user confirmed. Because the harmful action can happen long after the malicious input, the user may never see the link between the two, which makes the problem hard to notice and important to study.

The work sits in agentic memory systems and follows a continual learning paradigm: the LLM stays frozen while new attack patterns keep appearing, so the guard combines adversarial and contrastive learning with verified offline updates.

2 Research Problem

- **What is the problem?** Persistent memory poisoning in which the harm appears only when a stored record is retrieved in a later context. In a compositional attack several acceptable records become harmful together; in a dormant attack a later query supplies the trigger [8]. Most defenses decide too early, before the later query, related memories and intended action are known.
- **Who is affected?** Users and organisations that deploy memory-enabled agents for email, scheduling, customer service, web tasks or healthcare, and the developers responsible for those agents. The cost falls on the user when an agent leaks data, approves an action or follows a false preference.
- **Why does the problem exist?** Memory separates the moment of storage from the moment of use. Summarisation removes the original source, retrieval places unrelated records side by side, and the LLM tends to treat retrieved text as instructions. Summaries and tool echoes can also make an untrusted claim look more reliable than its true origin [10].
- **What limits current approaches?** Prompt guards score one string at a time [4]; memory-native consensus needs several LLM calls [5]; unknown-attack graph detectors model agents rather than memories [6]; and authority controls block laundered permissions but do not judge content [10].

- **Why is further research needed?** A write-time consistency check cut overall compromise from 62.55% to 20.09% but remained weak on compositional and dormant cases [8], and storage, harmful action and repair turn out to be different outcomes [9]. No compact defense has been tested across all of them.

Problem statement Can context-dependent memory risk be detected at practical cost while the system keeps an immutable record of origin and a separate authorisation check for consequential actions?

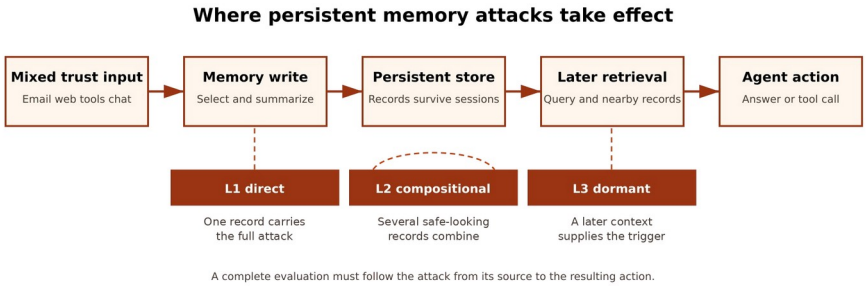


Figure 1 Attack points across the persistent memory lifecycle

3 Research Gap

What existing studies have addressed. Attack studies have shown that memory can be poisoned both with direct store access and through ordinary queries [2], [3]. Benchmarks have extended this to enterprise channels [7], to multi-record and delayed attacks [8] and to the full path from storage to action and repair [9]. Defenses exist at single points in the pipeline: before storage [4], during reasoning [5], after multi-agent communication [6] and at the final action [10].

What limitations remain. Each defense sees only part of the context. PIGuard does not see the later query, related memories or source. A-MemGuard sees memory context but roughly doubles token use and assumes most retrieved evidence is honest. BlindGuard detects unfamiliar attacks, but its nodes are agents rather than memory records. TMA-NM depends on correct source labels and does not identify harmful content.

What has not been adequately investigated. No reviewed study combines a compact contextual detector with authenticated provenance, tests it explicitly on compositional and dormant attacks and on held-out attack families, and measures write, retrieval, action, repair, utility and cost under one controlled setup.

How this research addresses the gap. CPMG scores a memory with its later query or action, related memories and immutable origin, at write and at retrieval time, using a single small encoder rather than repeated LLM calls. A controlled MPBench extension supplies compositional and dormant cases, and the evaluation separates the learned detector from the deterministic authority policy. Origin-bound authority is used as an existing control; the contribution is the detector and its evaluation.

4 Research Questions

RQ1 Does conditioning on the later query or action, neighbouring memories and authenticated provenance reduce compositional and dormant attack success compared with pointwise detection at matched false-positive rates?

RQ2 Does corruption-guided contrastive training with verified rehearsal generalise to unseen attack families and channels without forgetting earlier families or increasing benign false positives?

RQ3 How does the proposed approach compare with PIGuard, A-MemGuard and an adapted BlindGuard in security, clean utility and cost, and how much additional protection comes from the origin-bound action policy?

5 Research Objectives

1. To review and compare twelve studies on agent memory, memory attacks, benchmarks and defenses, and document the gap they leave (supports all RQs).
2. To build a controlled extension of MPBench with at least 150 scenario groups balanced across direct, compositional and dormant attacks and at least five channels, each paired with a benign counterfactual (RQ1).
3. To develop CPMG, a compact cross-encoder conditioned on memory, source interaction, current query or action, up to five related memories and provenance, trained with classification, contrastive and rehearsal objectives (RQ1, RQ2).
4. To measure end-to-end attack success, TPR at 1% and 5% FPR and AUROC on held-out attack families, channels and domains, and change on earlier families after adaptation (RQ1, RQ2).
5. To compare CPMG with the three baselines under the same agent, LLM, memory backend and attack budget, reporting LongMemEval accuracy, latency, tokens and LLM calls, with and without the authority policy (RQ3).

6 Literature Review

Twelve studies were selected from the NeurIPS, ICML, ACL and ICLR proceedings, the ACL Anthology, Google Scholar and arXiv. Eight were published in 2025 or later, and MemGPT and LoCoMo serve as foundational work. Eight are peer-reviewed (seven main-conference papers and one ICML workshop paper); the remaining four are recent arXiv preprints, included because they are currently the only work on compositional, dormant, lifecycle and authority-laundering poisoning, and their results are treated as provisional. The three baselines are all recent peer-reviewed methods: PIGuard (ACL 2025), A-MemGuard (ICML 2026) and BlindGuard (ACL 2026). Table 1 compares all twelve studies.

6.1 Memory Architectures

MemGPT manages working, recall and archival memory in the way an operating system manages main memory, and improved continuity on multi-session conversation and long-document tasks [1]. LoCoMo built very long multi-session conversations and showed that LLMs and RAG systems still lag humans at long-range temporal and causal recall [11]. LongMemEval turned this into 500 questions on extraction, multi-session reasoning, temporal reasoning, knowledge updates and

abstention, and found sharp accuracy drops in commercial assistants over sustained interaction [12]. All three establish why agents need persistent memory and provide the utility yardstick a defense must not damage, but none considers source trust or attack.

6.2 Attacks on Agent Memory

The attack literature has steadily reduced what the attacker needs. AgentPoison required write access to the store and an optimised trigger, yet with under 0.1% poisoned demonstrations reached about 81 to 82% retrieval success and 59 to 63% target action success with under 1% change in benign performance [2]. MINJA removed store access entirely, using ordinary queries and bridging steps to reach 98.2% injection success and 76.8% victim attack success, although its strongest settings assume shared or known memory [3]. Both show that clean-task accuracy cannot reveal a compromise.

6.3 Benchmarks

Three benchmarks widen the threat model in different directions. MPBench widens the input channels to files, websites, email, calendars, Slack, code and skills, and across 3,240 test cases on OpenClaw and HERMES reported 50.46% mean attack success [7]. MemPoison widens the attack structure into direct, compositional and dormant levels over 1,227 hand-validated cases; its write-consistency check lowered compromise from 62.55% to 20.09% but was much weaker on the last two [8]. MemSecBench widens the outcome from storage to action and repair, finding 84.2% persistence but only 50.3% end-to-end success and 56.1% selective repair [9]. Together they show that evaluation must follow a record past the point of storage.

6.4 Defense Methods

The four defenses act at different points and trade coverage against cost. PIGuard, a DeBERTa classifier trained to reduce over-defense, is fast at about 15 ms and reached 87.32% over-defense accuracy, but it sees one string at a time and transferred poorly to MPBench [4], [7]. A-MemGuard compares several reasoning paths over retrieved memories and stores failures in a lesson memory; it reduced direct and indirect attacks but used roughly twice the tokens and assumes most retrieved evidence is honest [5]. BlindGuard learns normal interaction patterns through directional corruption and contrastive learning and can flag unfamiliar attacks without labels, but its graph nodes are agents rather than memories [6]. TMA-NM binds origin at write time and raises authority only through trusted corroboration or user approval, reporting 0% success for laundering attacks in scope, but it relies on correct labels and does not rate content [10].

6.5 Synthesis

Across the twelve studies, three conclusions stand out. First, storage alone does not prove compromise; harm appears at retrieval and action, so a defense and its evaluation must reach those stages. Second, a record's risk can depend on the query and the other records retrieved with it, which a pointwise filter cannot see. Third, factual usefulness and authority are different properties: a low-authority source may inform an answer but should not approve an action. No existing method combines later context, related memories and authenticated origin without repeated LLM calls, and none has been evaluated explicitly on compositional and dormant attacks. These conclusions shape the proposed methodology.

Table 1 Literature Review Comparison of the Twelve Selected Papers

Paper	Problem	Method / approach	Dataset / data	Key findings	Limitations
MemGPT [1] arXiv 2023	Context window too small for long, multi-session tasks	OS-style virtual context with working, recall and archival tiers managed through function calls	Multi-session chat; document QA and nested key-value retrieval beyond the context window	Better long-range retrieval and cross-session consistency than fixed-context baselines	No threat model; stored content trusted regardless of source
LoCoMo [11] ACL 2024	Evaluating very long-term conversational memory	Human-verified generation of long multi-session dialogues with QA, summarisation and temporal tasks	Very long multi-session conversations (hundreds of turns each)	LLMs and RAG lag humans on long-range temporal and causal memory	Measures utility only; no adversarial setting
LongMemEval [12] ICLR 2025	Benchmarking long-term interactive memory of chat assistants	Scalable benchmark with five core memory abilities and a unified memory-design framework	500 curated questions embedded in long chat histories	Sharp accuracy drop for commercial assistants and long-context LLMs over sustained interaction	Utility only; no attacker model
AgentPoison [2] NeurIPS 2024	Backdooring an agent through its memory or knowledge base	Optimised trigger that maps triggered queries to poisoned demonstrations	Driving, knowledge-intensive QA and healthcare EHR agents	Under 0.1% poison: about 81 to 82% retrieval and 59 to 63% target action success; benign change under 1%	Direct store access; trigger needed in the later query
MINJA [3] NeurIPS 2025	Poisoning memory without store access	Query-only injection with bridging steps and progressive shortening	Healthcare, web and question-answering agents	98.2% injection success; 76.8% victim attack success	Shared or known victim memory; repeated turns
PIGuard [4] ACL 2025	Prompt-injection guards over-defend on benign trigger words	DeBERTa classifier trained with Mitigating Over-defense for Free	NotInject over-defense set and prompt-injection benchmarks	83.48% average accuracy; 87.32% over-defense accuracy; about 15 ms	Pointwise and content-only; weak transfer on MPBench
A-MemGuard [5] ICML 2026	Poisoned memory misleading agent reasoning	Multiple reasoning paths, consensus-divergence check and lesson memory	Direct and indirect memory-attack settings on LLM agents	Reduced both direct and indirect attack success	About 2x tokens; repeated LLM calls; assumes trustworthy majority
BlindGuard [6] ACL 2026	Unknown attacks in multi-agent systems without attack labels	Hierarchical graph encoder, directional corruption and contrastive learning	Multi-agent communication graphs under several attack types	Detects unfamiliar attacks without labeled examples of them	Nodes are agents, not memories; detects after communication
MPBench [7] ICML Wksp 2026	Exposure of real memory stacks across enterprise channels	Systematic attacks through four memory-write channels and six attack classes	3,240 malicious and benign test cases, six attack classes, seven domain types, on OpenClaw and HERMES	50.46% mean attack success; 41.05% retrieval success	Two agents and one LLM backend
MemPoison [8] arXiv 2026	Structural blind spots of isolated filtering	L1 direct, L2 compositional and L3 dormant taxonomy; write-consistency check	1,227 hand-validated cases, three memory substrates, ten model families	62.55% undefended; 20.09% with write check, weak on L2 and L3	Recent preprint; artifact availability uncertain
MemSecBench [9] arXiv 2026	Whether stored poison leads to action and can be repaired	Lifecycle benchmark over write, execution and forgetting	24 agent and memory configurations	84.2% persistence; 50.3% end-to-end success; 56.1% selective repair	Single trials; LLM-judge noise
TMA-NM [10] arXiv 2026	Laundering of untrusted authority through summaries and echoes	Origin binding, taint propagation and gated authority with machine-checked guarantees	Cross-defense, cross-attack benchmark over eight frontier models	Existing defenses up to 68% laundering success; TMA-NM 0% with full legitimate utility	Needs correct labels and independent principals; no semantic risk

7 Proposed Research Methodology

7.1 Research Approach

The research is **experimental**, combined with design and development of a prototype. The questions are comparative and causal: whether adding context, provenance or contrastive training lowers attack success. Answering them requires changing one factor, the defense, while holding the agent, LLM, memory backend, prompts and attack budget fixed, and measuring outcomes on the same cases. A survey could capture opinions about agent security but not attack success, and a design alone could not test whether it works better than existing methods.

7.2 Data Source

The study mainly uses **secondary data** from published benchmarks, plus a controlled extension that we construct ourselves (Table 2).

Table 2 Data Sources

Dataset	Type	Characteristics	Role
MPBench [7]	Secondary	3,240 malicious and benign test cases, six attack classes, seven domain types, two agents	Primary training, validation and test data
CPMG-L2L3 extension	Primary (constructed)	At least 150 scenario groups, balanced across direct, compositional and dormant attacks, each with a benign counterfactual	Main test of RQ1
AgentPoison [2], MINJA [3]	Secondary	Optimised backdoor and query-only injection attacks	Transfer to other attack types
NotInject [4]	Secondary	Benign text containing suspicious trigger words	Hard negatives against over-defense
LongMemEval [12]	Secondary	500 long-term memory questions covering extraction, multi-session and temporal reasoning	Clean utility only

7.3 Data Collection

Public datasets will be downloaded from their official releases and every case will be replayed through one instrumented harness that logs each memory write, retrieval, tool call and resulting tool state; these **system logs** are the primary measurements. The extension will be built from a stratified subset of MPBench by splitting instructions across records (compositional) or moving the trigger into a later query (dormant), then creating a matched counterfactual that keeps the vocabulary but removes the harmful relation. Deterministic checks confirm each case, and labels that cannot be checked in code are reviewed blind. Splits are grouped by scenario and entity to avoid leakage, and whole attack families, one channel and one domain are held out.

7.4 Proposed Method

CPMG is a deep learning classifier that sits around the existing memory pipeline (Figure 2). Every candidate memory arrives with an authenticated envelope recording content, channel, source,

time and parent tool call, which the LLM cannot edit. A DeBERTa-v3-base encoder receives the memory, its source interaction, the current query or planned action and up to five related memories, with special tokens for channel, authority and record age. It runs at write time to choose trusted storage, untrusted storage, quarantine or rejection, and again at retrieval time to rescore the memory in context. Training combines weighted cross-entropy, supervised contrastive learning on attack paraphrases and hard benign examples, BlindGuard-style synthetic corruptions of authority and relations, and offline rehearsal with a replay buffer when new verified attacks appear. Separately, a deterministic policy allows, asks for confirmation, asks for corroboration or denies each protected action based on the origin of its authorisation [10].

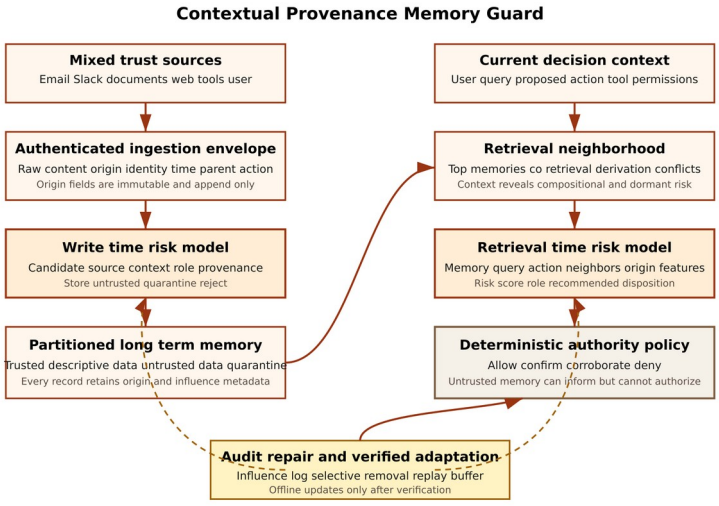


Figure 2 Architecture of the Contextual Provenance Memory Guard

Baselines are PIGuard, A-MemGuard and BlindGuard adapted to a graph of memory records, plus no-defense and origin-policy-only controls. All actions run in a sandbox with no real accounts or private data.

8 Evaluation Plan

Evaluation uses quantitative technical metrics at each stage of the memory lifecycle (Table 3), because the reviewed benchmarks show that storage, action and repair can diverge [9]; a single accuracy figure would hide where a defense succeeds or fails.

Table 3 Evaluation Metrics

Stage	Measure	Why it is appropriate
Write and retrieval	Poison storage and activation rate	Shows whether malicious content entered usable context
Action	End-to-end attack success rate and harmful tool state	Captures the real consequence for the user
Detection	AUROC, macro-F1, TPR at 1% and 5% FPR	Deployed guards must work at low false-positive budgets
Utility	LongMemEval accuracy and false rejection	Checks that useful memory is preserved
Efficiency	p50 and p95 latency, tokens and LLM calls	Tests the cost claim against A-MemGuard
Generalisation and continual learning	Held-out family, channel and domain deltas; old-family change after adaptation	Tests RQ2 on unseen attacks and forgetting

Thresholds are set on validation data only. Results use three seeds with bootstrap 95% confidence intervals; McNemar's test compares paired outcomes, with Holm correction across the three baselines. Ablations remove provenance, later context, related memories, contrastive training and the action policy in turn.

9 Expected Outcomes

We expect the following, stated as hypotheses rather than results. CPMG should reduce compositional and dormant attack success more than PIGuard at the same false-positive rate, because it sees the later query, planned action and related memories. Provenance features and contrastive corruption training should improve performance on unseen channels and attack families. As a single small encoder, CPMG should use far fewer LLM calls and tokens than A-MemGuard while achieving comparable or lower attack success, and verified rehearsal should limit forgetting when a new attack family is added. The authority policy should block unauthorised protected actions within the threat model even when the detector misses a record.

10 Significance of the Research

- **Who benefits** Developers and organisations deploying memory-enabled assistants, security researchers, and end users whose data those agents can reach.
- **Academic contribution** A context-conditioned formulation of memory risk, a compositional and dormant benchmark extension, and a fair lifecycle comparison of four defense types.
- **Practical contribution** A low-cost memory guard for Raymond and similar agents that learns from mixed-trust sources without giving them lasting authority, and supports selective removal of compromised records.
- **How it addresses the problem** It moves the decision to where context is known and separates content risk from action authority.

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